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MEE 214 — Past Questions & Materials
by Renance

Course: MEE 214
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 200

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Paper 1 of 6 — MEE 214 - 2024/2025 - Second Semester - Engineering Materials

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
 2024/2025 Academic Session, Second Semester Examination
COURSE CODE AND TITLE: GET 202/ MEE 214 – Engineering Materials.
DATE: September 23rd, 2025. TIME ALLOWED: 2 Hours
INSTRUCTIONS: Answer four (4) questions only.

- (1.a) With the aid of a tree diagram classify engineering materials. [4 Marks]
- (1.b) State the seven crystal lattice systems and give example each of the element/compound that adopts their crystal system [4 Marks]
- (1.c) A cylindrical alloy steel rod was prepared as a specimen for a tensile test before it was further used for the production of automobile universal shaft. If the specimen was loaded with 350N weight and subjected to a stress of 56 MPa. Calculate the diameter of the specimen. [9½ Marks] 2.82×10^{-3}
- (2.a) What is heat treatment. [4 Marks]
- (2.b) State seven (7) objectives of heat treatment. [4 Marks]
- (2.c) List and write a brief note on five (5) heat treatment processes commonly employed in engineering practice. [9½ Marks]
- (3.a) With the aid of a neat sketch draw the three crystal lattice usually found in most of the metals. [6 Marks]
- (3.b) The unit cell of chromium is cubic and contains 5 atoms. Calculate the lattice constant in terms of the radius. Given that $C_r = 52g$, Avogadro Number (AN) = 6.02×10^{23} . Density of chromium = $7.2 Mg/m^3$. [9½ Marks]
- (4.a) A binary alloy consisting of 40% A and 60% B, and at a certain temperature, have two phases, α (with 30% B) and β (with 70% B), in equilibrium. Determine the fractions of each phase. [9½ Marks]
- (4.b) State the three main types of binary phase diagram.
- (4.c) State some applications of Phase diagram. [4 Marks].
- (5.a) Discuss the effect of impurities on cast iron. [8 Marks]
- (5.b) Below is the Fe-C phase diagram, calculate the phases present at the eutectoid composition line at, temperature of 1538°C, 1200°C and 750°C.

Handwritten notes on the right side of the diagram: **all FCC**, **FeCr BCC**, **ZnMg HCP**.

Handwritten notes on the left side: **Shear-strain yield point.**

Handwritten notes at the bottom: **hardness**, **corrosion resistance**, **ductility**, **ability to deform before fracture**, **toughness**, **brittleness**, **fatigue**, **ability to fracture without deforming**, **malleability**, **ability to be hammered to shape**, **hexagonal closed**, **failure under repeated loading**.

[9½ Marks]

Paper 2 of 6 — MEE 214 2016 2017 Past Question

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
 First Semester Examination 2016/2017 Academic Session

COURSE CODE AND TITLE: MEE 214 - ENGINEERING MATERIALS
TIME ALLOTTED: TWO (2) HOURS
INSTRUCTION: ANSWER QUESTIONS ONE (1) ANY OTHER TWO

QUESTION 1

a. i. What is binary phase diagram *from composition*

ii. As shown on the phase diagram in fig. Q1 b, the alloy is at a temperature of 920 °C. Given that the overall composition C_0 is 23.7. (i) What phases are present? (ii) What are their exact compositions at this temperature? **10 Marks**

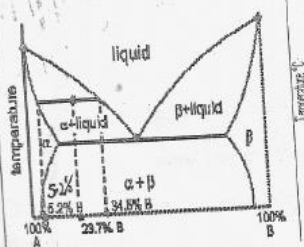


Fig. Q1 b.

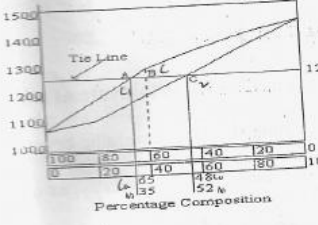


Fig. Q1 c.

b. Find the ratio of phases that exist at a temperature of 1268°C for a 60% Copper - 40% Nickel alloy, respectively as shown in fig. Q1c. Which phase has the highest proportion **10 Marks**

c. (i) A certain material is confirmed to obey Hook's law of elasticity, show that the deformation of the material, $\delta = PL/AE$, where P = load, L = length, A = cross sectional area and E = Young modulus

(ii) In an experiment, a circular steel specimen of 15 mm diameter was found to elongate by 0.2mm in a 200mm gauge length when it was subjected to a tensile force of 30KN. Determine the value of Young modulus for the steel, if the tested specimen did not exceed elastic limit. **10 Marks**

	C_1	C_2
% Cu	65	60
% Ni	35	40

	C_1	C_2
% Cu	65	60
% Ni	35	40

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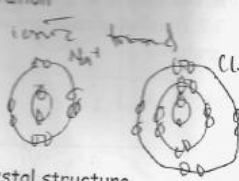
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QUESTION 2

- Differentiate between material science and engineering materials
- On which basis would you select materials for engineering purpose
- With the aid of well labeled diagram classify engineering materials according to their properties or uses

Q3 What type of bonding exists in sodium chloride and write down its electronic configuration

ionic bond



QUESTION 3

- What is crystal structure
- State the seven (7) crystal systems and give one example each
- Briefly explain the three metallic crystal structure you know
- Define corrosion and state two types of corrosion control techniques

TOC ARM
PCC
B
HCP
Zinc

QUESTION 4

State two types, properties and uses of the following Engineering materials

- Steel
- Cast iron
- Plastics
- Fiber reinforced polymer composites

- Cathodic protection
- Anodic protection

Iron magnetic
Impurities

Cargal Vanic
20 Marks
Clastic
Clastic

Suck

Soln

Using modulus (E) = $\frac{\text{Stress}}{\text{Strain}}$
= $\frac{F/A}{\Delta L/L}$
F = 30 kN = 30,000 N
= 15 mm = 0.015
A = 10.2 mm = 0.0002
= 200 mm = 0.2
A = $\pi r^2 = \pi (10.2 \times 10^{-3})^2$
= 3.27×10^{-4}
E = $\frac{30,000}{\frac{0.015}{3.27 \times 10^{-4}}}$
= 6.72×10^{11}
= $6.72 \times 10^{11} \text{ N/m}^2$



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Paper 3 of 6 — MEE 214 2014 2015 Past question

Level 200 • First Semester • PastQuestion

UNIVERSITY OF

DEPARTMENT OF. a

First Semester" Examinatiam ai4/ 2075 Session

COURSE CODE AND TITLE: MBE-214 - ENGINEERING MATERIALS

TIME ALLOTTED: 2 HOURS >

INSTRUCTION: ANSWER QUESBEONS ONE (1) ANY OTHER TWO

at a temperature of 1268°C for a 60% Copper - 40% Nickel

c and non-eutectic binary phase

line, solidus line

QUESTION 1

(a) Find the ratio of phases that exist

alloy. Which phase has the highest proportion (6) Draw a eutecti

diagram indicating the following regions: liquid, a +liquid, B+ liquid, a, B, a+P, liquidus

Jo — cuvpge

Se hicc — riers,

a MO th OL

Fig. Qt Sod i : , BOMARKS

QUESTION 7

(a.) Explain the following metallic c

Cubic (BCC) Structure (ii.) Face Cent

Structure (b.) state four (4) Difference between

ystal structures and give 2 examples each (i,) Body-Centered

tered Cubic (FCC) Structure ili. Hexagonal Close Packed (HCP)

Crystalline Solids and Amorphous Soltds.

' 20 MARKS

4

QUESTION 3

a. Differentiate between Adaterials Science and Engineering Materials. (b)With the aid of a

diagram classify engineering materials according to their properties or uses. {c) Differentiate

the following properties: of materials; (i) Resilience and toughness (ii) ductility and brittleness

(iii) malleability and machinability (iv) Creep and fatigue 20 MARKS

QUESTION 4

2. Define and derive an expression for the following; i. tensile stress and strain (ii) compressive

in. b. what is viscoelasticity ¢. A load of SKN is to be

stress and strain iii. Shear stress and stra

raised with the help of a steel wire. Find the minimum diameter of the steel wire, if the stress is
: 20 MARKS
not to exceed 100 MPa.

QUESTION 5 |

ween thermoset and thermoplastic materials. Give two (2) examples each

(a) Differentiate bet

(b) what is solid solution (c.) Explain two (2) types of solid solution 20 MARKS

Paper 4 of 6 — MEE214 LIKELY QUESTIONS Past Questions

Level 200 • First Semester • PastQuestion

MEE214LIKELYQUESTIONS

1. WHAT IS CRYSTALLOGRAPHY: crystal

lography is a branch of science

that deals with the formation, structure

and properties of crystals

and their applications

of crystals

and their properties

and their applications

of crystals

2. WHAT IS CRYSTAL STRUCTURE: A CRYSTAL STRUCTURE IS DEFINED

AS THE PARTICULAR REPEATING ARRANGEMENT OF ATOMS

THROUGHOUT THE CRYSTALS

3. WHAT IS SOLID: A SOLID IS ONE THAT HAS CHARACTERISTICS BY ITS

STRUCTURE, RIGIDITY AND RESISTANCE TO CHANGE OF SHAPE OR

VOLUME

4. TYPES OF SOLIDS: (

i) CRYSTALLINE SOLIDS (ii

) AMORPHOUS SOLIDS

5. WHAT IS A CRYSTALLINE SOLID: IT IS A WELL-ORDERED DEFINITE,

ARRANGEMENT OF ATOMS, MOLECULES, AND IONS

6. WHAT IS AN AMORPHOUS SOLID: AMORPHOUS SOLIDS OR NON-

CRYSTALLINE SOLIDS ARE SOLIDS THAT LACK THE LONG-RANGE

ORDER CHARACTERISTICS OF A CRYSTAL

7. DIFFERENCE BETWEEN CRYSTALLINE SOLIDS AND AMORPHOUS

SOLIDS

CRYSTALLINE SOLIDS AMORPHOUS SOLIDS

THEY HAVE CHARACTERISTICS

STI

C DOES NOT HAVE A DEFINITE

TE

GEOMETRIC

CSHAPE GEOMETRIC

CALSHAPE

THEY HAVE HIGHLY ORDERED PARTICLES

ARE HIGHLY

GHLY

THREE-

DIMENSIONAL ARRANGEMENT OF THREE DIMENSIONAL SOLIDS
 THEY HAVE SHARP MELTING POINTS AND THEY DON'T HAVE SHARP BOILING POINTS
 THEY HAVE SHARP MELTING POINTS
 EXAMPLES INCLUDES EXAMPLES INCLUDES COAL, COKE, COPPER SULPHATE (CuSO_4), NI, SO₄, GLASS, PLASTICS, RUBBER ETC.
 DIAMOND, GRAPHITE, NACL ETC.
 8. WHAT ARE THE WAYS TO IDENTIFY THE TYPE OF BONDING THAT HOLDS THE SOLID TOGETHER? (I) IONIC
 (C) SOLID
 (DS(ii)) MOLECULAR SOLIDS (ii)
 (i) METALLIC SOLID
 (DS(I)) NETWORK OR COVALENT SOLIDS.
 9. WHAT ARE IONIC SOLIDS? IONIC SOLID
 (DS, THE SOLID) CONSISTS OF DISCRETE CHEMICAL SPECIES HELD TOGETHER BY IONIC
 INTERMOLECULAR FORCES THAT ARE ELECTROSTATIC COULOMB IN NATURE.

10. EXPLAIN THE TERM MOLECULAR SOLIDS:

THE SOLID CONSISTS OF

STIFF

DISCRETE CHEMICAL SPECIES HELD TOGETHER BY INTERMOLECULAR FORCES THAT ARE ELECTROSTATIC COULOMB IN NATURE

11. WHAT ARE METALLIC SOLIDS: IN METALLIC SOLIDS THE VALENCE ELECTRONS ARE NO LONGER EXCLUSIVELY ASSOCIATED WITH A SINGLE ATOM. INSTEAD, THESE ELECTRONS EXIST IN MOLECULAR ORBITALS THAT ARE DELOCALIZED OVER MANY ATOMS, PRODUCING AN ELECTRONIC BAND STRUCTURE.

12. NETWORK SOLIDS: IN

NETWORK SOLIDS THERE ARE NO DISTINCT

MOLECULES OR CLUSTERS WITHIN THE STRUCTURE.

15. BRIEFLY EXPLAIN THE THREE CRYSTAL STRUCTURES YOU KNOW (I)

BODY-CENTERED CUBE (BCC):

THIS HAS ATOMS ARRANGED SUCH THAT

THEIR CENTERS ARE POSITIONED ON THE CORNERS OF A CUBE WITH ONE ATOM IN THE CENTER.

(i)

i)

.FACE-CENTERED CUBIC (FCC): HAS ATOMS LOCATED AT THE

CORNERS OF THE CUBE WITH ONE ATOM EACH AT THE CENTER OF EACH CUBIC FACE.

(ii)

i)

.HEXAGONAL CLOSE PACKED (HCP) STRUCTURE: IN

THIS

STRUCTURE, ATOMS ARE LOCATED AT THE 12 CORNERS OF A

HEXAGONAL PRISM WITH TWO AT THE CENTER OF THE TWO FACES OF THE PRISM AND THREE AT THE CENTER OF THE BODY.

16. WHAT IS LATTICE? A LATTICE IS A MATHEMATICAL

ABSTRACTION THAT DESCRIBES THE WAY THE ATOMS OR GROUPS OF ATOMS ARE REPEATED IN SPACE.

17. WHAT IS A UNIT CELL? A UNIT CELL IS THE SMALLEST GROUP OF ATOMS, I

ONS OR MOLECULES THAT, WHEN REPEATED AT REGULAR

INTERVALS IN THREE DIMENSIONS WILL PRODUCE THE LATTICE OF A CRYSTAL SYSTEM.

18. NOW WE HAVE 7 CRYSTAL LATTICE SYSTEMS AND 14 BRAVAILL

LATTICES.

19. LIST THESE SEVEN CRYSTAL LATTICE SYSTEMS AND GIVE ONE

EXAMPLE EACH

TRICLINIC K_2SO_8

MONOCLIC

As_4S_4 ,

KNO_3

RHOMBOHEDRAL Hg,

Sb

HEXAGONAL Zn,

Co,

NiAs

ORTHORHOMBIC

C Ga;

Fe_3C

TETRAGONAL Au,

Si,

NaCl

CUBIC

C NaCl

19. WHAT IS CRYSTALLISATION? CRYSTALLISATION IS THE

TRANSITION FROM THE LIQUID

TO THE SOLID STATE AND OCCURS IN

TWO STAGES.

20. WHAT ARE THE STAGES OF CRYSTALLISATION? (I)

(I) NUCLEUS

FORMATION (II)

(II) CRYSTAL GROWTH

21. WHAT IS BRAVAIS LATTICE? THE BRAVAIS LATTICES ARE THE BASIC

BUILDING BLOCKS FROM WHICH CRYSTALS CAN BE

CONSTRUCTED.

22. DEFINE PHASE. A PHASE IS A REGION OF SPACE THROUGHOUT

WHICH ALL PHYSICAL PROPERTIES OF MATERIAL ARE ESSENTIALLY

UNIFORM.

23. STATE THREE TYPES OF PHASES KNOWN (I)

(I) SOLID (II)

(II) LIQUID (I)

(I)

GASEOUS STATE

24. THERE ARE TWO TYPES OF PHASE MATERIAL, NAMELY PHASE

EQUILIBRIUM, AND REACTION EQUILIBRIUM

BRIUM

25. WHAT IS THE PHASE EQUILIBRIUM? REFER TO THE TRANSPORT OF MATTER BETWEEN PHASES OF THE SYSTEM WITHOUT CONVERSION OF ONE SPECIES TO ANOTHER.

26. WHAT IS A PHASE DIAGRAM OR THERMAL EQUILIBRIUM DIAGRAM (T-E)? A PHASE DIAGRAM IS A GRAPHICAL WAY TO REPRESENT THE VARIOUS PHASES OF A SUBSTANCE AND THE CONDITIONS UNDER WHICH EACH PHASE EXISTS.

27. STATE SOME OF THE APPLICATIONS OF PHASE DIAGRAM. (i) TO PREDICT THE ALLOY MICROSTRUCTURE (i

i

) A CRUCIAL PART OF METALLURGY (i

ii

) TO PREDICT THE BEHAVIOUR OF MATERIALS SUCH AS METALS.

28. DEFINE A BINARY SYSTEM. AND GIVE SOME EXAMPLES OF BINARY SYSTEM. A PHASE DIAGRAM THAT PLOTS TEMPERATURE AGAINST THE RELATIVE CONCENTRATIONS OR COMPOSITIONS OF TWO SUBSTANCES IS A BINARY MIXTURE. SOLID SOLUTION, EUTECTIC OR PERITECTIC ETC. ?

29. LIST THE THREE MAIN TYPES OF BINARY PHASE DIAGRAMS. THEY ARE COMPLETE SOLID SOLID SOLUTION, EUTECTIC DIAGRAM, AND PERITECTIC DIAGRAM.

30. WHAT IS LEVER RULE AND TIE LINE? LEVER RULE IS A RULE APPLIED IN IDENTIFYING THE PROPORTIONS OF PHASES PRESENT IN A PHASE DIAGRAM. WHEN A TIE LINE IS DRAWN THROUGH A POINT.

31. WHAT IS ATOMIC DIFFUSION? ATOMIC DIFFUSION IS A DIFFUSION PROCESS WHEREBY THE RANDOM THERMALLY-ACTIVATED MOVEMENT OF ATOMS IN A SOLID RESULTS IN THE NET TRANSPORT OF ATOMS.

33. WHAT IS SELF-DIFFUSION? SELF-DIFFUSION IS A DIFFUSION IN ONE COMPONENT MATERIAL, WHEN ALL ATOMS THAT EXCHANGE POSITIONS ARE OF THE SAME TYPE. EXAMPLES ARE THE

I

INTERDIFFUSION THAT OCCURS IN THE ALLOY OF COPPER AND NICKEL.

34. WHAT IS VISCOSITY? VISCOSITY IS A MEASURE OF THE RESISTANCE OF A FLUID TO THE SHEAR OR TENSILE STRESS.

35. WHAT IS AN IDEAL FLUID OR INVISCID FLUID? IS A FLUID THAT

DOESNOT HAVE RESISTANCE TO SHEAR STRESS.

36. STATE NEWTON'S LAW OF VISCOSITY? STATE THAT FOR STRAIGHT, PARALLEL AND UNIFORM FLOW, THE SHEAR STRESS BETWEEN LAYERS IS PROPORTIONAL TO THE VELOCITY GRADIENT (SHEAR STRAIN) IN THE DIRECTION PERPENDICULAR TO THE LAYERS.

$\frac{\partial y}{\partial x}$

1

.e. $\tau = \mu$

$\frac{\partial x}{\partial y}$

37. LIST THE TYPES (OR FORMS) OF VISCOSITY. NEWTONIAN, SHEAR THICKENING, SHEAR THINNING, THIXOTROPIC, RHEOPECTIC, A BINGHAM PLASTIC, AND A MAGNITUDO-RHEOLOGICAL FLUID.

38. WHAT ARE METALS? A METAL IS A CHEMICAL ELEMENT THAT IS A GOOD CONDUCTOR OF BOTH ELECTRICITY AND HEAT AND FORMS CATIONS AND IONIC BONDS WITH NON-METALS.

39. LIST FOUR (4) CHARACTERISTICS OF METALS.

i

. THEY ARE STRONG, AND CAN BE FORMED EASILY INTO USEFUL SHAPES.

ii

. THEIR SURFACES HAVE A CHARACTERISTIC METALLIC LUSTRE.

i

ii

. THEY ARE GOOD CONDUCTORS OF ELECTRIC CURRENT AND HEAT.

i

v. THEY CAN BE DEFORMED, ALLOWING THEM TO WITHSTAND SUDDEN AND SEVERE LOADS.

40. LIST OUT THE IMPORTANT MECHANICAL PROPERTIES OF METALS: THESE INCLUDE STRENGTH, STIFFNESS, ELASTICITY, PLASTICITY, DUCTILITY, BRITTLENESS, MALLEABILITY, TOUGHNESS, RESILIENCE, CREEP AND HARDNESS.

41. WHAT IS TENSILE TESTING (TENSION TESTING)? IT IS A FUNDAMENTAL MATERIAL SCIENCE TEST IN WHICH A SAMPLE IS SUBJECTED TO A TENSILE FORCE UNTIL FAILURE OCCURS.

42. WHAT ARE THE PROPERTIES THAT ARE DIRECTLY MEASURED WITH A (VI

A) TENSILE TEST

1. ULTIMATE TENSILE STRENGTH

1

I. MAXIMUM ELONGATION

II

I. ANDREDUCTIBILITY AREA.

43. WHAT ARE THE USES OF THE RESULTS FROM THE TENSILE TEST?

i. THEY ARE USED TO SELECT MATERIAL FOR AN APPLICATION

i

i. ARE USED FOR QUALITY CONTROL

i

i

i. AND TO PREDICT HOW A MATERIAL WILL BEHAVE UNDER TYPES OF FORCES.

44. WHAT ARE THE TWO TYPES OF UNIVERSAL MACHINES WE HAVE.

i. HYDRAULIC POWERED

i

i. ELECTROMAGNETIC POWERED MACHINE

45. WHAT ARE THE CAPABILITIES OF A GOOD UNIVERSAL TESTING MACHINE

i. FORCE CAPACITY

i

i. SPEED

i

i

i. PRECISION AND ACCURACY

46. WHAT IS A LOAD? A LOAD IS DEFINED AS ANY EXTERNAL FORCE ACTING UPON A MATERIAL OR MACHINE PART.

47. LIST THE WAYS IN WHICH A LOAD CAN BE APPLIED TO A MATERIAL.

i. DEAD OR STEADY LOAD: THE LOAD DOES NOT CHANGE IN MAGNITUDE OR DIRECTION

i

i. LIVE OR VARIABLE LOAD: A LOAD IS VARIABLE WHEN IT CHANGES CONTINUALLY

i

i

i. IMPACT LOAD: IT IS APPLIED WITH SOME INITIAL VELOCITY

i

v. SUDDENLY APPLIED OR SHOCK LOADS: A LOAD IS SAID TO BE SUDDENLY APPLIED OR SHOCK LOAD, WHEN IT IS SUDDENLY APPLIED OR REMOVED

48. WHAT IS STRESS? THESIS WHEN SOME EXTERNAL SYSTEM OF FORCES OR LOADS ACT ON A BODY, THE INTERNAL FORCES (EQUAL AND OPPOSITE) ARE SET UP AT VARIOUS SECTIONS OF THE BODY,

WHICH RESIST THE EXTERNAL FORCES.

49. LI

STHE TWO TYPES OF FORCES ACTING ON MATERIALS AND EXPLAIN.

i

. TENSILE STRESS: THIS IS A RESULT OF WHEN A BODY IS SUBJECTED TO TWO EQUAL AND OPPOSITE PULLS

ii

. TENSILE STRAIN: THIS IS A RESULT WHEN A TENSILE LOAD IS APPLIED TO A BODY THERE IS AN INCREASE IN THE CROSS-SECTIONAL AREA AND AN INCREASE IN THE LENGTH OF THE MATERIAL

50. WHAT IS STRESS? THIS IS WHEN SOME EXTERNAL SYSTEM OF FORCES OR LOADS ACT ON A BODY, THE INTERNAL FORCES ARE SET UP AT VARIOUS SECTIONS OF THE BODY, WHICH RESIST THE EXTERNAL FORCES.

51. WHAT IS STRAIN? THIS IS WHEN SOME EXTERNAL FORCES OR LOADS ACT ON A BODY, IT UNDERGOES SOME DEFORMATION.

52. WHAT IS A STRESS-STRAIN DIAGRAM? A STRESS-STRAIN DIAGRAM IS A DIAGRAM THAT EXPRESSES A RELATIONSHIP BETWEEN A LOADS APPLIED TO THE MATERIAL, CAUSED BY THE LOAD.

53. WHAT IS TORSIONAL SHEAR STRESS? THIS IS WHEN A MATERIAL OR MACHINE PART IS SUBJECTED TO THE ACTION OF TWO EQUAL AND OPPOSITE COUPLES ACTING IN PARALLEL.

54. THE HARDNESS OF A METAL MAY BE DETERMINED BY.

i

. BRINELL HARDNESS TEST

ii

. ROCKWELL HARDNESS TEST

i

ii

. VICKERS HARDNESS (ALSO DIAMOND PYRAMID) TEST AND

i

v. SHORE SCLEROSCOPE.

55. HOW CAN CAST IRON BE OBTAINED? CAST IRON CAN BE OBTAINED BY REMELTING PIG IRON WITH COKE AND Limestone in a furnace known as a cupola.

56. LI

STOUT THE PROPERTIES OF CAST IRON.

i

. ITSCOSTI SLOW

ii

. GOODCASTI NGCHARACTERI STICS

i

ii

. HIGHCOMPRESSI VESTRENGTH

i

v. WEARRESI STANCEAND

v. EXCELLENTMACHI NABI LITY.

57.WHATARETHEUSESOFCASTI RON?

i

. ITCANBEUSEDTOMAKEMACHI NETOOLBODIES.

ii

. INMARI NEENGINEERINGTHEYAREUSEDFORPLASTI C
REINFORCEMENTOFBOATHALLS

i

ii

. ITISUSEDI NMAKINGGLASSPANELSOFAI RCRAFTS

i

v. ITISUSEDFORTHERMALI NSULATIONNOFMACHI NEPARTS

58.LI

STOUTTHETYPESOFCASTI RONS.

i

. GREYCASTI RON

ii

. WHITECASTI RON

i

ii

. CHILLEDCASTI RON

i

v. MOTTLEDCASTI RON

v. MALLEABLECASTI RON

vi

. NODULARORSPHEROI DALGRAPHI TECASTI RON

59.WHATAREFERROUSMETALSANDNONFERROUSMETALS?

i

. FERROUSMETALS: ARETHOSEWHI CHHAVEI RONSASTHERE
MAI NCONSTI TUENT.

ii

. NONFERROUSMETALS: ARETHOSEWHI CHCONTAI NAMETAL

OTHERTHANIRONASTHEIRCHEMICALFUNCTIONS.

60.WHYARENONFERROUSMETALSUSUALLYEMPLOYEDIN
INDUSTRY

i

. EASEOFFABRICATION(CASTING, ROLLING, FORGING, WELDING,
ANDFABRICATION)

ii

. RESISTANCETOCORROSION

i

ii

. ELECTRICALANDTHERMALCONDUCTIVITYAND

i

v. WEIGHT

61.WHATARETHEIRPURITIESFOUNDINCASTIRONANDWHATARE
THEIREFFECTS?

i

. SILICON

CON: IT

TPROVIDESTHEFORMATIONOFFREEGRAPHITE

WHICHMAKESTHEIRONSOFTANDEASILYMACHINABLE,IT
ALSOPRODUCESSOUNDCASTINGSFREEFROMBLOWHOLES,
BECAUSEOFITSHIGHAFFINITYFOROXYGEN

ii

. SULPHUR: IT

TMAKESTHECASTIRONHARDANDBRITTLE

i

ii

. MANGANESE: IT

TMAKESTHECASTIRONWHICH

TEANDHARD,

ANDIT ALSOHELPEXERTACONTROLLINGINFLUENCEOVER
HARMFULEFFECTSOFSULPHUR.

i

v. PHOSPHOROUS: ITAIDSFUSIBILITY

ANDFLUIDITYINCASTIRON.

62.HOWISWROUGHTIRONPRODUCED?ITISPRODUCEDFROMPIG
IRONBYREMELTINGTIN THEPUDDLINGFURNACEOFA
REVERTORYTYPE.

63.LIST

THE TYPES OF STEEL.

i

. ALLOYSTEEL: MAYBEDEFI NEASSTEELTOWHI CHELEMENT OTHERTHANCARBONAREADDEDI NSUFFI CIENTAMOUNTTO PRODUCEANI MPROVEMENTI NPROPERTI ES.

ii

. CARBONSTEEL: ISDEFI NEDASTHESTEELWHI CHHASI TS PROPERTI ESMAI NLYDUETOI TSCARBONCONTENT.

i

ii

. STAI NLESSSTEEL: ITISDEFI NEDASTHESTEELWHI CHWHEN CORRECTLYHEATEDANDFI NISHED, RESISTSOXI DATIONAND CORROSIONATTACKFROM MOSTCORROSI VEMEDI A

i

v. HEATRESI STI

NGSTEEL: ARESTEELSWHI CHCANRESI STCREEP ANDOXI DATIONATHI GHTEMPERATURESANDRETAI N SUFFICI ENTSTRENGTH.

v. SPRI NGSTEEL: ARETHEMOSTSUI TABLEMATERI ALFOR SPRINGSANDCANSTOREUPMAXI MUM AMOUNTOFWORKOR ENERGYI NAGI VENWEI GHTORVOLUMEOFSPRI NGMATERI AL, WITHOUTPERMANENTDEFORMATI ON.

vi

. HI GHSPEEDTOOLSTEELS: THESESTEELSAREUSEDFOR CUTTINGMETALSATAMUCHHI GHERCUTTI NGSPEEDTHAN ORDINARYCARBONTOOLSTEELS.

64.WHATARETHEUSESOFSTEEL

i

. ITI SUSEDFORMAKI NGPI PEFI TTINGS,DOORHI NGES.

ii

. ITI SUSEDTODESI GNPRODUCTSTHATREQUI RESHI GH RESISTANCETOFRI CTIONSUCHASCARWHEELS, JAW CRUSHER.

i

ii

. MOSTSTEELAREUSEDFORAI R-PLANEFUSELAGEAND AUTOMOBILEPARTS

65.DEFINETHETERM HEATTREATMENTOFSTEELS.THETERM HEATTREATMENTOFSTEELMAYBEDEFI NEASANOPERATI ONORA COMBI NATIONOFOPERATI ONS, INVOLVI NGTHEHEATI NGAND COOLINGOFAMETALORANALLOYI NTHESOLI DSTATEFORTHE PURPOSEOFOBTAI NINGCERTAI NDESI RABLECONDI TIONSOR

PROPERTIES WITHOUT CHANGE IN TEMPERATURE.

66. WHAT IS THE AIM OF HEAT TREATMENT?

i.

. TO INCREASE THE HARDNESS OF METALS

ii.

. TO RELIEVE THE STRESSES SET UP IN THE MATERIAL AFTER
HOT OR COLD WORKING

i.

ii.

. TO IMPROVE MACHINABILITY

i.

v. TO SOFTEN THE METAL

v. TO MODIFY THE STRUCTURE OF THE MATERIAL TO IMPROVE

i.

TO SELECTRIC AND MAGNETIC PROPERTIES

vi.

. TO CHANGE THE GRAIN SIZE

vii.

. TO INCREASE THE QUALITY OF A METAL

67. LIST

THE VARIOUS HEAT TREATMENTS COMMONLY EMPLOYED IN
ENGINEERING PRACTICE.

i. NORMALISING

i.

i. ANNEALING

i.

i.

i. HARDENING

i.

v. TEMPERING

v. SPHEROIDISING

NG

v.

i. SURFACE HARDENING OR CASE HARDENING

68. LIST

THE NON-FERROUS METALS.

i. ALUMINIUM

i.

i. COPPER

i.

i.

i. LEAD

i

v. TI N

v. ZI NC

v

i. NI CKELANDTHEI RALLOY

69.THEBASI CMANUFACTURI NGPROCESSESI NENGINEERINGCAN
BECATEGORI ZEDI NTOFIVETYPESNAMELY.

i

. CASTI NG

ii

. METALFORMI NG

i

ii

. MACHI NING

i

v. JOI NING

v. ANDFI NISHINGPROCESS

70.WHATARECERAMI CS?CERAMI CSAREI NORGANI CANDNON-
METALLI CMATERI ALSTHATARECOMMONLYUSEDFORELECTRI CAL
ANDTHERMALI NSULATORSANDAREBRI TTLEINNATUREWHI CH
COMPOSEOFMORETHANONEELEMENT.

71.WHATARETHECHARACTERRI STICSOFCERAMI CSMATERIALS?

i

. THEYAREABRASI VERESISTANT,HEATRESI STANCE
(REFRACTORY)ANDCANSUSTAI NLARGECOMPRESSI VELOADS
EVENATHI GHTEMPERATURES

ii

. MANYCERAMI CSARECHEMI CALINERTEVENATHI GH
TEMPERATURESASI SSHOWNBYTHEI RGOODOXI DATIONAND
REDUCTI ONRESISITANTATTHESETEMPERATURES

i

ii

. CERAMI CSAREINORGANI C,NON-METALLI CMATERIALSTHAT
AREPROCESSEDAND/ ORUSEDATHI GHTEMPERATURES
ANDARECOMMONLYELECTRI CALANDTHERMALI NSULATORS,
COMPOSEDOFMORETHANONEELEMENT

i

v. THEYHAVEBEENSUBJECTEDHEATTREATMENT

72.LI

STOUTTHEVARI OUSTYPESOFCERAMI CSMATERIALS

ACCORDI NG TO THE I RMETHOD OF PRODUCTI ON?

i. WHI TEWARE WHI CHI NCLUDES CHI NA AND PROCELAI N

i

i. STRUCTURAL CLAY PRODUCTS

i

i

i. GLASS AND

i

v. REFRACTORY MATERI ALS,WHICH ARE CAPABLE OF

WI THSTANDI NG VERY HI GH TEMPERATURES

73.LI

STTHE EXAMPLES OF CERAMI C MATERI ALS.

i. ALL TYPE S OF GLASS PRODUCTS

i

i. ABRASI VE SANDS AND SOME TYPE S OF CUTTI NG TOOLS

i

i

i. BRICKS, TILE,SEWER AND DRAI N PIPE

i

v. MANY TYPE S OF ELECTRI CAL INSULATI

ONS, FERRO- ELECTRICS,

FERRO MAGNETI C AND SEMI CONDUCTORS

v. REFRACTORI ES FOR HI GH TEMPERATURE USE

v

i. PORCELAI N ENAMELS AND REFRACTORY COATI NGS FOR

METALS

vi

i. DINNER- WARE, SANITORY WARE, ARTWARE ETC.

74.LI

STTHE VARI OUS CERAMI CS FORMI NG TECHNI QUES.

i. HYDROPLASTI C FORMI NG

i

i. SLIPCASTI NG

i

i

i. DRYI NG AND FI RI NG

i

v. POWDER PRESSI NG

v. CEMENTI TIOUS BOND

v

i. INJECTI ON MOLDI NG

75. WHAT IS GLASS? A GLASS IS A TRANSPARENT SOLID PRODUCT WHICH MAY BE AMORPHOUS OR CRYSTALLINE AND DEPENDS UPON HEAT TREATMENT.

76. WHAT ARE THE MAJOR STEPS IN PRODUCING GLASS PRODUCTS?

i. MELTING AND REFINING

i

ii. FORMING AND SLEETING

i

i

iii. HEAT-TREATMENT AND

i

v. FINISHING

77. WHAT ARE THE FOUR METHODS USED TO FABRICATE GLASS?

i. PRESSING

i

ii. BLOWING

i

i

iii. DRAWING

i

v. FIBERFORMING

78. LIST

THE TWO HEAT-TREATMENTS OF GLASS MATERIAL.

i. ANNEALING

i

ii. TEMPERING

79. LIST

THE TYPES OF GLASS.

i. SODA-LIME GLASSES

i

ii. LEAD GLASSES

i

i

iii. BOROSILICATE GLASSES

i

v. ALUMINOSILICATE GLASSES

v. FUSED-SILICA

GLASSES

80. WHAT ARE THE USES OF GLASS?

i

i. Indoors, windows and furnitures

ii

. LABORATORY EQUIPMENTS AND CHEMICAL GLASSWARE

i

ii

. DOMESTIC UTENSILS

i

v. OPTICAL GLASSES

v. RADIOACTIVE VALVES

vi

. GLASS CURTAIN WALL ON HIGH RISE BUILDING

NG

vii

. ELECTRICAL INSULATORS

vi

ii

. GLASS PIPE, FIBRE GLASS INSULATOR, GLASS SPRINGS

i

x. AS THERMAL INSULATOR

x. X- RAY TUBE

81. WHAT ARE REFRACTORIES?

REFRACTORIES ARE CERAMIC MATERIALS THAT CAN WITHSTAND USUALLY HIGH HEAT AS WELL AS ABRASION AND THE CORROSIVE EFFECTS OF ACIDS AND ALKALIS.

82. WHAT ARE THE USES OF REFRACTORIES?

i

. THEY ARE USED IN FURNACE

ii

. FOR THE CRACKING OF PETROLEUM, CERAMIC LINSING, BOILER,

INSULATOR

ETC.

i

ii

. JET ENGINES USE REFRACTORY PARTS WITHOUT IT, JET ENGINES DO NOT LAST VERY LONG.

83. WHAT ARE THE CHARACTERISTICS

OF REFRACTORIES?

REFRACTORIES?

i

. REFRACTORIES ARE HEAT, CORROSION AND ABRASION RESISTANT

ii

. THEY POSSESS HIGH FUSION TEMPERATURE

i

ii

. THEY POSSESS LOW THERMAL COEFFICIENT OF EXPANSION

i

v. THEY DO NOT HAVE CHEMICAL AFFINITY WITH THE MOLTEN METAL THEY HOLD

v. THEY WITHSTAND HIGH TEMPERATURES AND PRESSURE

vi

. THEY SHOULD POSSESS HEAT AND ELECTRICAL

INSULATING

PROPERTIES

84. THE CLASSIFIED

CATEGORIES OF REFRACTORIES ARE

i

. ACID REFRACTORIES

ii

. BASIC REFRACTORIES

i

ii

. NEUTRAL REFRACTORIES

85. THE INDUSTRIES WHERE CONSUMERS OF REFRACTORIES ARE

i

. IRON AND STEEL

ii

. NON-FERROUS METALS

i

ii

. STEAM AND POWER

i

v. OIL REFINING

v. GLASS AND POTTERY

vi

. CEMENT LIME AND OTHER MINERALS

vii

. CHEMICAL INDUSTRIES

86. WHAT IS AN ABRASIVE MATERIAL? AN ABRASIVE IS A HAND,

MECHANICALLY RESISTANT MATERIAL USED FOR GRINDING

AND CUTTING.

87. LIST

THE TWO TYPES OF ABRASIVE MATERIALS

ALS.

i. NATURALASI N(DI AMOND, GARNET, FLINT ETC.)

i

i. SYNTHETIC CEMENT (SILICON CARBIDE AND ALUMINIUM OXIDE)

88. WHAT IS CEMENT? CEMENT IS A SUBSTANCE USED AS BINDER, FOR CONSTRUCTION WHERE CHISELS, HARDEN AND ADHERE TO OTHER MATERIALS.

89. LIST

THE TYPES OF CEMENT.

i

. NATURAL CEMENT

ii

. ARTIFICIAL CEMENT (PORTLAND CEMENT)

i

ii

. QUICK SETTING CEMENT

i

v. RAPID HARDENING CEMENT

v. SULPHATE RESISTING CEMENT

vi

. LOW HEAT CEMENT

vii

. HIGH ALUMINA CEMENT

vi

ii

. COLOURED CEMENT

i

x. BLAST FURNACE SLAG CEMENT

x. HYDROGRAPHIC CEMENT

90. LIST

THE TWO TYPES OF POLYMERS.

i. NATURAL POLYMER

i

i. SYNTHETIC POLYMER

91. WHAT ARE THE CHARACTERISTICS OF POLYMERS?

i. LOW DENSITY

i

i. LOW COEFFICIENT OF FRICTION

i

i

i. GOODMOULDABILITY

i

v. CANBEPRODUCEDWITHCLOSEDIMENSIONS

ONALTOLERANCES

v. EXCELLENTSURFACEFINISHCANEObtained

v

i. POORTENSILESTRENGTH

vi

i. POORTEMPERATURESTRENGTH

92.WHATISPOLYMERIZATION?

I

STHEPROCESSOFCONVERTINGAMONOMERORAMIXTUREOF

MONOMERSINTOAPOLYMER.

93.WHATARETHETHREEGENERALMETHODSORMECHANISMSOF

POLYMERISATION?

i. ADDITION

ONPOLYMERIZATION

i

i. COPOLYMERIZATION

i

i

i. CONDENSATIONPOLYMERIZATION

94.WHATISAPLASTICMATERIAL?

PLASTICSARETHEFINALPRODUCT, CREATEDAFTERONEORMORE

POLYMERSORADDITION

VEHAVEBEENADDEDTOA RESISTING

PROCESSING,WHICHISTHENSHAPEDINTOAFINALFORM.

95.LI

STOUTTHETYPESOFPLASTICS.

i. THERMOSETTING

i

i. THERMOPLASTICS

96.WHATARETHEPROPERTIESOFPLASTICS?

i. THEYARELESSBRITTLETHANGLASS

i

i. THEYRESISTCORROSIONANDACTIVELYOFCHEMICALS

i

i

i. THEYARELIGHTWEIGHTANDPOSSESGOODSTRENGTH

i

v. THEY HAVE HIGH DIELECTRIC STRENGTH HENCE SUITABLE FOR
ELECTRIC INSULATION

97. THE USES OF PLASTICS ARE.

i. IT IS USED FOR MAKING DISPOSABLE COMMODITY PLASTICS

i

i. IT CAN BE USED TO REPLACE METALS IN SOME PARTS OF
MACHINES

i

i

i. IT CAN BE USED TO MAKE INSULATOR

98. WHAT IS THE DIFFERENCE BETWEEN THERMOSETTING AND
THERMOPLASTIC

THERMOSETTING RESINS THERMOPLASTIC

i

. ONCE HARDENED AND THEY CAN BE REPEATEDLY
SET, THEY DO NOT SOFTEN BY HEAT AND
SOFTEN WITH THE HARDENED BY COOLING

APPLICATIONS OF
HEAT.

i

i

. THEY ARE STRONGER THEY ARE COMPARATIVELY
AND HARDER THAN SOFTER AND LESS STRONG
THERMOPLASTIC

C

RESINS

OBJECTS MADE BY
THERMOPLASTIC RESINS

i

i

i. OBJECTS MADE BY
CANNOT BE USED AT
THERMOSETTING
NG
COMPARATIVELY HIGHER
RESINS CAN BE USED AT TEMPERATURES AS THEY WILL
AT COMPARATIVELY TEND TO SOFTEN UNDER HEAT

HIGHER
TEMPERATURE
WITHOUT DAMAGE.

i
v. THEY ARE USUALLY SUPPLIED AS
EDAS
SUPPLIED IN GRANULAR MATERIAL.
MONOMER OR
PARTIALLY
POLYMER SED FORM

I
N WHICH THEY ARE
EITHER
THERMALLY SET OR
PARTIALLY
THERMOPLASTIC
SOLIDS

99. THERMOSETTING PLASTICS CAN BE FORMED BY.

i. COMPRESSION OR TRANSFER MOLDING

i

i. CASTING

100. THERMO- PLASTICS CAN BE FORMED BY.

i. INJECTION MOLDING

i

i. BLOW MOLDING

i

i

i. EXTRUSION

i

v. CALENDERING

v. THERMO- FORMING

v

i. CASTING

101. WHAT ARE ELASTOMERS (RUBBERS)

AN ELASTOMER IS A POLYMERIC MATERIAL THAT MAY EXPERIENCE
LARGE AND REVERSIBLE ELASTIC DEFORMATIONS.

102. WHAT ARE THE CHARACTERISTICS

OF RUBBERS?

i. THEY ARE ESSENTIALLY NON-CRYSTALLINE IN STRUCTURE

i

i. THEY ARE NON CONDUCTORS OF ELECTRICITY AND ARE RELATIVELY LOW HEAT CONDUCTORS.

i

i

i. THEY HAVE RELATIVELY LOW SOFTENING TEMPERATURES

i

v. THEY ARE HIGHLY RESISTANT TO CHEMICALS AND RESISTANCE TO CHEMICAL CORROSION IN ENVIRONMENT

v. THEY EXHIBIT

SCHEMATIC BEHAVIOUR GENERALLY TO A

GREATER EXTENT THAN PLASTICS.

103. WHAT ARE THE PROPERTIES OF ELASTOMERS?

i

. RESILIENCE AND HYSTERESIS

i

i

. HARDNESS

ii

i

. TENSILE STRENGTH

iv

. COMPRESSION SET OR CREEP

v

. TEAR RESISTANCE

vi

. ABRASION RESISTANCE

vii

i

. DETERRIORATION RESISTANCE

viii

i

. OIL RESISTANCE

104. LIST THE TYPES OF ELASTOMERS OR RUBBERS

i

. NATURAL RUBBER

i

i

. STYRENE-BUTADIENE RUBBER (SBR)

ii

i

. CHLOROPHENE (NEOPRENE) (CR)

iv

. ISOBUTYLENE- ISOPRENE(BUTYL)(UR)

v

. ISOPRENE

vi

. POLYACRYLATE(ACM)

v i

i

. NI TRI LR BUTADI ENE(NBR)

vii

i

. POLYBUTADI ENE(BR)

i

x. POLYSULFI DE(THI OLCOL)

x. ETHYLENEPROPYLENE(EPM ANDEPDM)

xi

. URETHANE(AUANDEU)

xii

. CHLOROSULFONLYPOLYETHYLENE(HYPALON)(
CSM)

FLUOROCARBON(FKM)

xii

i

. SI LICONE

105.WHATI SVULCANI SATIONOFRUBBER?

VULCANI SATI ONI STHEI RREVERSI BLECHEMI CALREACTI
ON

I

NVOLVI NGSULPHUROROTHERSUI TABLEAGENTWHEREI N
CROSSLINKSAREFORMEDBETWEENMOLECULARCHAIN SINRUBBER
MATERIALS.

106.LISTTHEFORMINGANDFABRICATINGTECHNIQUESFOR
RUBBERS

i

. COMPRESSIONMOULDING

i

i

. CALENDERING

ii

i

. EXTRUSION

iv

. INJECTI ONANDTRANSFERMOULDI NG

v

. VACUM FORMI NGOFSHEETRUBBER

vi

. CASTI NGOF LI QUI D(URETHANEANDPOLYSULPHI DE)RUBBERS

107.THEFI NISHI NGMETHODSFORRUBBERSARE

i. TUMBLI NG, EITHER DRYORI NLI QUI DCARBON-DI

OXIDE TO

REMOVE FLASH

i

i. DIETRI MMI NG

ii

i. TEARTRI MMI NG

iv. BUFFI NG

108.WHATARETHE USES OF RUBBER?

i

. VEHI CLE TYRES

i

i

. ERASERS

ii

i

. GASKETS

iv

. GLOVES AND APRONS

v

. BELT CONVEYORS

vi

. SEALS

vi

i

. O- RINGS

vii

i

. WATERSTOPS

i

x. HOSES

x. FLOORTI LES

xi

. ADHESI VESANDTAPES

xii

. EXPANSI ONJOI NTS

xii

i

. CORROSI ONRESI STANTCOATI NGS,LINNI NGS

xiv

. PI PESANDCONTAI NERS

xv. WI REANDCABLECOVERI NGS

xvi

. VI BRATIONMOUNTSI NAUTOMOBILE

xv i

i

. AUTOMOBILEUPHOLSTERY

xvii

i

. ENGINEMOUNTI NGS

109.WHATARECOMPOSITEMATERIALS?

AREMADEOFDIFFERENTMATERIALSTHATAREFORCED

TOGETHER(EXAMPLE: FIBER

GLASS, CONCRETE, ANDWOOD)TO

ACHIEVESPECIFICPROPERTIES.

110.LISTTHETYPESOFFIBER-REIN

FORCEDCOMPOSITES.

i

. SYNTHETICAND

i

i

. NATURAL

111.WHATARETHEUSESOFFIBER-REIN

FORCEDCOMPOSITES?

i

. THEYAREUSEDTOREPLACEALUMINUMALLOY,WHICHPROVIDE

HIGHSTRENGTHANDFAIRLYHIGHSTIFFNESSATLOWWEIGHT.

i

i

. AREUSEFORSTRUCTURALAPPLICATIONS

121.WHATARETHEPROPERTIESOFFIBER-REINFORCEDCOMPOSITE

i

. THEYHAVEHIGHTENSILESTIFFNESS, ANDSTRENGTH

i

i

. HAVEVERYSMALLBENDI NGSTIFFNESS

ii

i

. THEYAREHARDBUTBRI TTLEAPATITE

122.WHATPROPERTI ESDOCOMPOSITIONSDDEPENDON?

i. PROPERTIESOFPHASES

i

i. GEOMETRYOFDI SPERSEDPHASE(

PARTI

CLES

ZE,

DI

STRIBUTION,

ANDORIENTATI

ON)

i

i

i. AMOUNTOFPHASE.

Paper 5 of 6 — MEE 214 ENGINEERING MATERIALS 2023,2024 Past Questions.

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

First'Semester Examination 2023/2024 Session

COURSE CODE AND TITLE: MEE 214 - ENGINEERING MATERIALS

TIME ALLOTTED: 2 HOURS

INSTRUCTION: ANSWER FOUR QUESTIONS. AT LE

SECTION B

SECTION A

Q1 (a) Differentiate between crystalline and amorphous structures in material

(4 Marks). Provide an example of materials with each type of structure (2 Marks) (b) Describ 4 crystal structures, such as vacancies, interstitials, and dislocations (5.5 Marks), Discuss their impact on material

properties and behaviour (6 Marks).

AST TWO FROM SECTION A AND AT LEAST ONE FROM

Is in terms of structure and properties

e common defects in

and insulators based on their electronic

plications (6 Marks). (b) Explain the term

dded to iron to

Q2 (a) Explain the classification of materials as conductors, semiconductors,

structure (6 Marks), Provide an example of each type of material and their ap

alloying and state the primary purpose of alloying (3.5 Marks). Give the primary alloying element a produce (i) stainless steel (1 Mark). (ii) cast iron (1 Mark).

Q3 (a) For a 40 wt% Sn-60 wt% Pb_alloy at 150°C shown in Fig. Q3 below: (i) what phase(s).is (are) present? (2 Marks)

(ii) What is (are) the composition(s) of the phase(s)? (10 Marks)

Fig. Q3

b) Make a simple sketch of fig. Q3 above (1.5 Marks) and identify the following on your sketch; a) solidus line b)

Liquidus line (2 Marks). c): Invariant point. (2 Marks) he

SECTION .B oe

Q4

(8 Marks). (b) A steel bar 12.7mm in diameter supports a load

a) Discuss the elastic deformation of metals and alloys

Tf the bar has a modulus of elasticity of 2.05 X

of 7000Kg)(i) What is the stress placed on the bar? (4 Marks) (ii)

10⁶ Pa, how-much will the bar be strained with the load? (5.5 Marks)

Qs

_ a) What is SHE? (2 Marks) (b) The potential difference of a new mined, dis

this could be achieved? (7.5 Marks) (c) Given that the standard energies of Fe and Cr³⁺ are -84.94 and -205 kJ/mol

respectively. Calculate E⁰ for Fe/Fe²⁺ and Cr/Cr³⁺ (8 Marks) (Take Faraday constant (F) to be 96540 C).

ly discovered metal is to be determined, discuss how

Paper 6 of 6 — MEE 214 PAST QUESTIONS

Level 200 • First Semester • PastQuestion

RESOURCES EFFURUN

ENGINEERING

2016/2017 Academic Session

ey

& MATERIALS

COURSE CODE AND TITLE: MEE 214

TIME ALLOTTED: TWO (2) HOURS ,

INSTRUCTION: ANSWER QUESTIONS 'ONE (4) ANY OTHER Two

ESTION 1 y

a. i, What is binary phase diagram

ii. As shown on the phase diagram in fig. QUB, the alloy is at a tem

Given that the overall composition Co's 23.7. (i)What phases are prese!

are their exact compositions at this temperature?

perature of 920°C.

nt? (ii)What

10 Marks

Calc |G

s pelele

jo copper

jwo}o0 = Nickle

35 sy

Percentage Carposition

Fig. Q1b. Fig. Qle.

Find the ratio of phases that exist at a temperature of 1268°C for a 60% Copper = 40%

Nickel alloy, respectively as shown in fig. Qic. Which phase has the highest proportion

10 Marks

ok's law of elasticity, show that the

material is confirmed to obey Hor

load, l= length, A= cross sectional area

c. (i) A certain

$\Delta l = \frac{Pl}{AE}$, where P =

deformation of the material,

and E = Young modulus

(ii) In an experiment, a circular steel specimen of 15 mt

'+h when it was subj

by 0.2mm in a 200mm gauge length
the value of Young modulus for the steel
mm diameter was found to elongate
subjected to a tensile force of

10 kN, if the tested specimen did

not exceed elastic limit. 5

Eu Cw» Cc> ee We 10 Marks

Io Cy 65 bo oe : Vv yw

10 Marks

10 Marks

Science and engineering materials

<? materials for engi

ineering purpose

diagram classify engineering materials according to

in sodium chloride and y

we

write down its electronic

20 Marks

a. What is crystal structure

“4

TOC AR Mm

b. State the seven (7) crystal systems and give one example each

c. Briefly explain the three metallic crystal structure you know

10 Marks

types of corrosion control techniques Here,

20 Marks

d. Define creep and state two

How the equation

Say 2 of

QUESTION 4 Get. 10 Marks

ee

State two types, Properties and uses of the following Engineering materials

a. Steel — Cth

b. Cast iron

c. Plastics

20 Marks

to : Gorge Vanie

Snlieaad

ay

Cheate

UNIVERSITY OF

DEPARTMENT OF.

First Semester Exe .

COURSE CODE AND TITLE: MEE 28% - ENGINEERING MATERTALS

TIME ALLOTTED: 2 HOURS >

INSTRUCTION: ANSWER QUESRIONS ONE (1) ANY OTHER TWO

QUESTION 1

(a) Find the ratio of phases that alloy. Which phase has the highest proportion (6) Draw a eu' dicating the following regions; liquid, a +liquid, B+ liquid, a, B, exist at a temperature of 1268°C for a 60% Copper - 40% Nickel tectic and non-eutectic binary phase diagram in a+®, liquidus line, solidus line

Fig. Qt CN | . : 30MARKS

QUESTION 7

(a.) Explain the: following metallic cryst Cubic (BCC) Structure (ii.) Face Centered Cubic Structure (.) state four (4) Difference between al structures and give 2 examples each (i,) Body-Centered (FCC) Structure iii. Hexagonal Close Packed (HCP) Crystalfine Solids and Amorphous Solids.

m 20 MARKS

QUESTION >

% i

a. Differentiate between Materials Science and Engineering Materials. (b)With the aid of « diagram classify engineering materials according to their properties or uses. (c) Differentiate the following properties: of materials; (i) Resilience and toughness (ii) ductility and brittleness (iii) malleability and mackinability, (iv) Creep and fatigue 20 MARKS

QUESTION 4

@. Define and derive an expression for the following: i. tensile stress and strain (ii) compressive nd strain. b. what is viscoelasticity ¢. A load of 5KN is to be stress and strain iii. Shear stress ai raised with the help of a steel wire. Find the minimum diameter of the steel wire, if the stress is 20 MARKS

not to exceed 100 MPa.

QUESTION 5

- (a) Differentiate between thermoset and thermoplastic materials.
 (b) what is solid solution i.e.) Explain two (2) types of solid-solution materials. Give two (2) examples each

20 MARKS

Ba

MEE214L1 KELYQUESTIONS

1. WHAT IS CRYSTALLOGRAPHY: crystallography is a branch of science that deals with the formation, structure, and properties of crystals.

2.

3.

4.

5.

6.

7.

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13.

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17.

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19.

20.

21.

22.

23.

24.

25.

26.

27.

28.

29.

30.

31.

CSHAPE GEOMETRI
 CALSHAPE
 THEYHAVEHIGHLYORDERED PARTI
 CLESAREHI
 GHLY
 THREE-
 DIMENI
 SIONAL ARRANGEDI
 NTHREEDIMENSI
 ONS
 ARRANGEMENTOFPARTICLES
 THEYHAVESHARPMELTI
 NGAND THEYDON'
 THAVESHARP
 BOI
 LINGPOI
 NTS MELTI
 NGPOINTS
 EXAMPLESINCLUDES EXAMPLESI
 NCLUDESCOAL, COKE,
 COPPERSULPHATE(CUSO₄)
 ,NI
 SO₄, GLASS,
 PLASTI
 C,RUBBERETC.
 DIAMOND,GRAPHITENACLETC.
 8.WHATARETHEWAYSOFI DENTIFYI
 NGTHETYPEOFBONDING
 THATHOLDSTHESOLI DTOGETHER? (I
)IONI
 C
 SOLI
 DS(ii
)MOLECULARSOLIDS(ii
 i
)METALLICSOLI
 DS(I
 V)NETWORK
 ORCOVALENTSOLIDS.
 9.WHATAREIONICSOLIDS?I
 NI ONICSOLI

DS, THE SOLID CONSISTS OF DISCRETE CHEMICAL SPECIES HELD TOGETHER BY INTERMOLECULAR FORCES THAT ARE ELECTROSTATIC COULOMB IN NATURE.

10. EXPLAIN THE TERM MOLECULAR SOLID: THE SOLID CONSISTS OF DISCRETE CHEMICAL SPECIES HELD TOGETHER BY INTERMOLECULAR FORCES THAT ARE ELECTROSTATIC COULOMB IN NATURE

11. WHAT ARE METALLIC SOLIDS: IN METALLIC SOLIDS THE VALENCE ELECTRONS ARE NO LONGER EXCLUSIVELY ASSOCIATED WITH A SINGLE ATOM. INSTEAD, THESE ELECTRONS EXIST IN MOLECULAR ORBITALS THAT ARE DELOCALIZED OVER MANY ATOMS, PRODUCING AN ELECTRONIC BAND STRUCTURE.

12. NETWORK SOLIDS: IN NETWORK SOLIDS THERE ARE NO DISTINCT MOLECULES OR CLUSTERS WITHIN THE STRUCTURE.

15. BRIEFLY EXPLAIN THE THREE CRYSTAL STRUCTURES YOU KNOW (I) BODY-CENTERED CUBE (BCC): THE SOLID HAS ATOMS ARRANGED SUCH THAT THEIR CENTERS ARE POSITIONED ON THE CORNERS OF A CUBE WITH ONE ATOM IN THE CENTER.

(i)

i) .FACE-CENTERED CUBIC (FCC): HAS ATOMS LOCATED AT THE CORNERS OF THE CUBE WITH ONE ATOM EACH AT THE CENTER OF EACH CUBIC FACE.

(ii)

i) .HEXAGONAL CLOSE PACKED (HCP) STRUCTURE: IN THIS STRUCTURE, ATOMS ARE LOCATED AT THE 12 CORNERS OF A HEXAGONAL PRISM WITH TWO AT THE CENTER OF THE TWO FACES OF THE PRISM AND THREE AT THE CENTER OF THE BODY.

16. WHAT IS LATTICE? A LATTICE IS A MATHEMATICAL ABSTRACTION THAT DESCRIBES THE WAY THE ATOMS OR GROUPS OF ATOMS ARE REPEATED IN SPACE.

17. WHAT IS A UNIT CELL? A UNIT CELL IS THE SMALLEST GROUP OF ATOMS, IN

ONSORMOLECULESTHAT, WHENREPEATEDATREGULAR
INTERVALSI NTHREEDI MENS I ONSWI LLPRODUCETHELATTICEOFA
CRYSTALSYSTEM.

18.N.BWEHAVE7CRYSTALLATTI CESYSTEM AND14BRAVAI S
LATTICES.

19.LISTTHESEVENCRYSTALLATTI CESYSTEM ANDGI VEONE
EXAMPLEEACH
TRINCLINC K₂S₂O₈

MONOCLI

NIC As₄S₄,

KNO₂

RHOMBOHEDRAL Hg,

Sb

HEXAGONAL Zn,

Co,

NiAs

ORTHORHOMBI

C Ga;

Fe₃C

TETRAGONAL Au,

Si,

NaCl

CUBI

C NaCl

19.WHATI SCRYSTALLISATI ON?CRYSTALLI SATI ONISTHE
TRANSI TIONFROM THELI QI

UDTOTHE SOLI DSTATEAND OCCURSI N
TWO STAGES.

20.WHATARETHESTAGESOFCRYSTALLI SATION?(I

)NUCLEUS

FORMATI ON(II

)CRYSTALGROWTH

21.WHATI SBRAVAI SLATTICE?THEBRAVAI SLATTI CEI STHEBASI C
BUILDINGBLOCKFROM WHI CHALLCRYSTALSCANBE
CONSTRUCTED.

22.DEFINEPHASEAPHASEI SAREGI ONOFSPACETHROUGHOUT
WHI CHALLPHYSI CALPROPERTI ESOFMATERI ALAREESSENTI ALLY
UNIFORM.

23.STATETHREETYPESOFPHASEKNOWN(I

)SOLI D(II

)LIQUID(I

II

GASEOUS STATE

24. THERE ARE TWO TYPES OF PHASE MATERIAL, NAMELY PHASE

EQUILIBRIUM, AND REACTION EQUILIBRIUM

BRUIM

25. WHAT IS PHASE EQUILIBRIUM? REFERS TO THE TRANSPORT OF MATTER BETWEEN PHASES OF THE SYSTEM WITHOUT CONVERSION OF ONE SPECIES TO ANOTHER

26. WHAT IS A PHASE DIAGRAM OR THERMAL EQUILIBRIUM DIAGRAM (T-X)? A PHASE DIAGRAM IS A GRAPHICAL WAY TO REPRESENT THE VARIOUS PHASES OF A SUBSTANCE AND THE CONDITIONS UNDER WHICH EACH PHASE EXISTS.

27. STATE SOME OF THE APPLICATIONS OF PHASE DIAGRAM. (i) TO PREDICT THE ALLOY MICROSTRUCTURE (i

i

) A CRUCIAL PART OF

METALLURGY (i

ii

) TO PREDICT THE BEHAVIOUR OF MATERIALS SUCH AS METALS.

28. DEFINE A BINARY SYSTEM. AND GIVE SOME EXAMPLES OF BINARY SYSTEM. A PHASE DIAGRAM THAT PLOTS TEMPERATURE AGAINST THE RELATIVE CONCENTRATION OR COMPOSITIONS OF TWO SUBSTANCES IS A BINARY MIXTURE. SOLID SOLUTION, EUTECTIC OR PERITECTIC ETC. ?

29. LIST THE THREE MAIN TYPES OF BINARY PHASE DIAGRAMS. THEY ARE COMPLETE SOLID SOLID SOLUTION, EUTECTIC DIAGRAM, AND PERITECTIC DIAGRAM.

30. WHAT IS LEVER RULE AND TIE LINE? LEVER RULE IS A RULE APPLIED IN IDENTIFYING THE PROPORTIONS OF PHASES PRESENT IN A PHASE DIAGRAM. WHEN A TIE LINE IS DRAWN THROUGH A POINT.

31. WHAT IS ATOMIC DIFFUSION? ATOMIC DIFFUSION IS A DIFFUSION PROCESS WHEREBY THE RANDOM THERMALLY-ACTIVATED MOVEMENT OF ATOMS IN A SOLID RESULTS IN THEIR NET TRANSPORT OF ATOMS.

33. WHAT IS SELF-DIFFUSION? SELF-DIFFUSION IS A DIFFUSION IN ONE COMPONENT MATERIAL, WHEN ALL ATOMS THAT EXCHANGE POSITIONS ARE OF THE SAME TYPE. EXAMPLES ARE THE

I

INTERDIFFUSION THAT OCCURS IN THE ALLOY OF COPPER AND NICKEL.

34. WHAT IS VISCOSITY? VISCOSITY IS A MEASURE OF THE RESISTANCE OF A FLUID WHICH SUBJUGATED BY EITHER SHEAR OR TENSIONAL STRESS.

35. WHAT IS AN IDEAL FLUID? IDEAL FLUID IS A FLUID THAT DOES NOT HAVE RESISTANCE TO SHEAR STRESS.

36. STATE NEWTON'S LAW OF VISCOSITY? STATE THAT FOR STRAIGHT, PARALLEL AND UNIFORM FLOW, THE SHEAR STRESS BETWEEN LAYERS IS PROPORTIONAL TO THE VELOCITY GRADIENT (SHEAR STRAIN) IN THE DIRECTION PERPENDICULAR TO THE LAYERS.

$\frac{\partial y}{\partial x}$

1

$\tau = \mu$

$\frac{\partial x}{\partial y}$

37. LIST THE TYPES (OR FORMS) OF VISCOSITY. NEWTONIAN, SHEAR THICKENING, SHEAR THINNING, THIXOTROPIC, RHEOPECTIC, A BINGHAM PLASTIC, AND A MAGNETO-RHEOLOGICAL FLUID.

38. WHAT ARE METALS? A METAL IS A CHEMICAL ELEMENT THAT IS A GOOD CONDUCTOR OF BOTH ELECTRICITY AND HEAT AND FORMS CATIONS AND IONIC BONDS WITH NON-METALS.

39. LIST FOUR (4) CHARACTERISTICS OF METALS.

i

1. THEY ARE STRONG, AND CAN BE FORMED EASILY INTO USEFUL SHAPES.

ii

2. THEIR SURFACES HAVE A CHARACTERISTIC METALLIC LUSTRE.

i

ii

3. THEY ARE GOOD CONDUCTORS OF ELECTRIC CURRENT AND HEAT.

i

4. THEY CAN BE DEFORMED, ALLOWING THEM TO WITHSTAND SUDDEN AND SEVERE LOADS.

40. LIST OUT THE IMPORTANT MECHANICAL PROPERTIES OF METALS: THESE INCLUDE STRENGTH, STIFFNESS, ELASTICITY, PLASTICITY, DUCTILITY, BRITTLENESS, MALLEABILITY, TOUGHNESS, RESILIENCE, CREEP AND HARDNESS.

41. WHAT IS TENSILE TESTING (TENSION TESTING)? IT IS A FUNDAMENTAL MATERIAL SCIENCE TEST IN WHICH A SAMPLE IS

SUBJECTED TO UNIAXIAL TENSION UNTIL FAILURE OCCURS.

42. WHAT ARE THE PROPERTIES THAT ARE DIRECTLY MEASURED

WITH A (VI

A) TENSILE TEST

I. ULTIMATE TENSILE STRENGTH

I

I. MAXIMUM ELONGATION

II

I. AND REDUCTION IN AREA.

43. WHAT ARE THE USES OF THE RESULTS FROM THE TENSILE TEST?

i. THEY ARE USED TO SELECT MATERIALS FOR AN APPLICATION

i

i. ARE USED FOR QUALITY CONTROL

i

i

i. AND TO PREDICT HOW A MATERIAL WILL BEHAVE UNDER TYPES OF FORCES.

44. WHAT ARE THE TWO TYPES OF UNIVERSAL MACHINES WE HAVE.

i. HYDRAULIC POWERED

i

i. ELECTROMAGNETIC POWERED MACHINE

45. WHAT ARE THE CAPABILITIES OF A GOOD UNIVERSAL TESTING MACHINE

i. FORCE CAPACITY

i

i. SPEED

i

i

i. PRECISION AND ACCURACY

46. WHAT IS A LOAD? A LOAD IS DEFINED AS ANY EXTERNAL FORCE

ACTING UPON A MATERIAL OR MACHINE PART.

47. LIST THE WAYS IN WHICH A LOAD CAN BE APPLIED TO A MATERIAL.

i. DEAD OR STEADY LOAD: THE LOAD DOES NOT CHANGE IN MAGNITUDE OR DIRECTION

i

i. LIVE OR VARIABLE LOAD: A LOAD IS VARIABLE WHEN IT CHANGES CONTINUALLY

i

i

i. IMPACT LOAD: IT IS APPLIED WITH SOME INITIAL VELOCITY

i

v. SUDDENLY APPLIED OR SHOCK LOADS: A LOAD IS SAID TO BE SUDDENLY APPLIED OR SHOCK LOAD, WHEN IT IS SUDDENLY APPLIED OR REMOVED

48. WHAT IS STRESS? THIS IS WHEN SOME EXTERNAL SYSTEM OF FORCES OR LOADS ACT ON A BODY, THE INTERNAL FORCES (EQUAL AND OPPOSITE TO THEM) ARE SET UP AT VARIOUS SECTIONS OF THE BODY, WHICH RESIST THE EXTERNAL FORCES.

49. LI

STHE TWO TYPES OF FORCES ACTING ON MATERIALS AND EXPLAIN.

i

. TENSILE STRESS: THIS IS A RESULT OF WHEN A BODY IS SUBJECTED TO TWO EQUAL AND OPPOSITE PULLS

ii

. TENSILE STRAIN: THIS IS A RESULT WHEN A TENSILE LOAD IS APPLIED TO A BODY THERE IS AN INCREASE IN THE CROSS-SECTIONAL AREA AND AN INCREASE IN THE LENGTH OF THE MATERIAL

50. WHAT IS STRESS? THIS IS WHEN SOME EXTERNAL SYSTEM OF FORCES OR LOADS ACT ON A BODY, THE INTERNAL FORCES ARE SET UP AT VARIOUS SECTIONS OF THE BODY, WHICH RESIST THE EXTERNAL FORCES.

51. WHAT IS STRAIN? THIS IS WHEN SOME EXTERNAL FORCES OR LOADS ACT ON A BODY, IT UNDERGOES SOME DEFORMATION.

52. WHAT IS A STRESS-STRAIN DIAGRAM? A STRESS-STRAIN DIAGRAM IS A DIAGRAM THAT EXPRESSES A RELATIONSHIP BETWEEN A LOADS APPLIED TO THE MATERIAL, CAUSED BY THE LOAD.

53. WHAT IS TORSIONAL SHEAR STRESS? THIS IS WHEN A MATERIAL IS PUT UNDER TORSION, IT IS SUBJECTED TO THE ACTION OF TWO EQUAL AND OPPOSITE COUPLES ACTING IN PARALLEL.

54. THE HARDNESS OF A METAL MAY BE DETERMINED BY.

i

. BRINELL HARDNESS TEST

ii

. ROCKWELL HARDNESS TEST

i

ii

. VICKERS HARDNESS (ALSO KNOWN AS PYRAMID) TEST AND

i

v. SHORE SCLEROSCOPE.

55. HOW CAN CAST IRON BE OBTAINED? CAST IRON CAN BE OBTAINED BY RE-MELTING THE PIG IRON WITH COKE AND LIMESTONE IN A FURNACE KNOWN AS CUPOLA.

56. LIST

THE PROPERTIES OF CAST IRON.

i

. IT IS COSTLY AND SLOW

ii

. GOOD CAST IRON CHARACTERISTICS

i

ii

. HIGH COMPRESSIVE STRENGTH

i

v. WEAR RESISTANCE AND

v. EXCELLENT MACHINABILITY.

57. WHAT ARE THE USES OF CAST IRON?

i

. IT CAN BE USED TO MAKE MACHINE TOOL BODIES.

ii

. IN MARINE ENGINEERING THEY ARE USED FOR PLASTIC REINFORCEMENT OF BOAT HULLS

i

ii

. IT IS USED IN MAKING GLASS PANELS OF AIRCRAFTS

i

v. IT IS USED FOR THERMAL INSULATION OF MACHINE PARTS

58. LIST

THE TYPES OF CAST IRONS.

i

. GREY CAST IRON

ii

. WHITE CAST IRON

i

ii

. CHILLED CAST IRON

i

v. MOTTLED CAST IRON

v. MALLEABLE CAST IRON

vi

. NODULAR OR SPHEROIDAL GRAPHITIC CAST IRON

59. WHAT ARE FERROUS METALS AND NON-FERROUS METALS?

i

. FERROUS METALS: ARE THOSE WHICH HAVE IRON IN THEM. MAJOR CONSTITUENT.

ii

. NON-FERROUS METALS: ARE THOSE WHICH CONTAIN METAL OTHER THAN IRON. IRON IS NOT A CONSTITUENT.

60. WHY ARE NON-FERROUS METALS USUALLY EMPLOYED IN INDUSTRY

i

. EASE OF FABRICATION (CASTING, ROLLING, FORGING, WELDING, AND FABRICATION)

ii

. RESISTANCE TO CORROSION

i

ii

. ELECTRICAL AND THERMAL CONDUCTIVITY AND

i

v. WEIGHT

61. WHAT ARE THE IMPURITIES FOUND IN CAST IRON AND WHAT ARE THEIR EFFECTS?

i

. SILICON

CON: IT

IMPROVES THE FORMABILITY OF FREEGRAPHITE

WHICH MAKES THE IRON SOFT AND EASILY MACHINABLE, IT

ALSO PRODUCES SOUND CASTINGS FREE FROM BLOWHOLES,

BECAUSE OF ITS HIGH AFFINITY FOR OXYGEN

ii

. SULPHUR: IT

MAKES THE CAST IRON HARD AND BRITTLE

i

ii

. MANGANESE: IT

MAKES THE CAST IRON WHICH

IS HARD,

AND IT ALSO HELPS EXERT A CONTROLLING INFLUENCE OVER

HARMFUL EFFECTS OF SULPHUR.

i

v. PHOSPHOROUS: IT AIDSFUSIBILITY

AND FLUIDITY IN CAST IRON.

62. HOW IS WROUGHT IRON PRODUCED? IT IS PRODUCED FROM PIG IRON BY REMELTING IN THE PUDDLE REFINERY TYPE OF A REVERBERATORY TYPE.

63. LIST

THE TYPES OF STEEL.

i

. ALLOY STEEL: MAY BE DEFINED AS STEEL TO WHICH SOME ELEMENT OTHER THAN CARBON ARE ADDED IN SUFFICIENT AMOUNT TO PRODUCE AN IMPROVEMENT IN PROPERTIES.

ii

. CARBON STEEL: IS DEFINED AS STEEL WHICH HAS ITS PROPERTIES MAINLY DUE TO ITS CARBON CONTENT.

i

ii

. STAINLESS STEEL: IT IS DEFINED AS STEEL WHICH WHEN CORRECTLY HEATED AND FINISHED, RESISTS OXIDATION AND CORROSION ATTACK FROM MOST CORROSIVE MEDIA.

i

v. HEAT RESISTENT

STEEL: ARE STEELS WHICH CAN RESIST CREEP AND OXIDATION AT HIGH TEMPERATURES AND RETAIN SUFFICIENT STRENGTH.

v. SPRING STEEL: ARE THE MOST SUITABLE MATERIALS FOR SPRINGS AND CAN STORE UP MAXIMUM AMOUNT OF WORK OR ENERGY IN A GIVEN VOLUME OF SPRING MATERIAL, WITHOUT PERMANENT DEFORMATION.

vi

. HIGH SPEED TOOL STEELS: THESE STEELS ARE USED FOR CUTTING METALS AT A MUCH HIGHER CUTTING SPEED THAN ORDINARY CARBON TOOL STEELS.

64. WHAT ARE THE USES OF STEEL

i

. IT IS USED FOR MAKING PIPES, DOORS, ETC.

ii

. IT IS USED TO DESIGN PRODUCTS THAT REQUIRE HIGH RESISTANCE TO FRICTIONS SUCH AS CAR WHEELS, JAW CRUSHER.

i

ii

. MOST STEEL ARE USED FOR AIR-PLANE FUSELAGE AND

AUTOMOBILE PARTS

65. DEFINE THE TERM HEAT TREATMENT OF STEELS. THE TERM HEAT TREATMENT OF STEEL MAY BE DEFINED AS AN OPERATIONAL OR A COMBINATION OF OPERATIONS, INVOLVING THE HEATING AND COOLING OF A METAL OR AN ALLOY IN THE SOLID STATE FOR THE PURPOSE OF OBTAINING CERTAIN DESIRABLE CONDITIONS OR PROPERTIES WITHOUT CHANGE IN TEMPERATURE.

66. WHAT IS THE AIM OF HEAT TREATMENT?

i

. TO INCREASE THE HARDNESS OF METALS

ii

. TO RELIEVE THE STRESSES SET UP IN THE MATERIAL AFTER HOT OR COLD WORKING

i

ii

. TO IMPROVE MACHINABILITY

i

v. TO SOFTEN THE METAL

v. TO MODIFY THE STRUCTURE OF THE MATERIAL TO IMPROVE

i

THE ELECTRICAL AND MAGNETIC PROPERTIES

vi

. TO CHANGE THE GRAIN SIZE

vii

. TO INCREASE THE QUALITY OF A METAL

67. LIST

THE VARIOUS HEAT TREATMENTS COMMONLY EMPLOYED IN ENGINEERING PRACTICE.

i. NORMALISING

i

i. ANNEALING

i

i

i. HARDENING

i

v. TEMPERING

v. SPHERODISING

NG

v

i. SURFACE HARDENING OR CASE HARDENING

68. LIST

STOUT THENON- FERROUS METALS.

i. ALUMINIUM

i

i. COPPER

i

i

i. LEAD

i

v. TIN

v. ZINC

v

i. NICKEL AND THE RALLOY

69. THE BASIC MANUFACTURING PROCESSES IN ENGINEERING CAN BE CATEGORIZED INTO FIVE TYPES NAMELY.

i

. CASTING

ii

. METAL FORMING

i

ii

. MACHINING

i

v. JOINING

v. FINISHING PROCESS

70. WHAT ARE CERAMICS? CERAMICS ARE INORGANIC AND NON-METALLIC MATERIALS THAT ARE COMMONLY USED FOR ELECTRICAL AND THERMAL INSULATORS AND ARE BUILT IN NATURE WHICH COMPOSE OF MORE THAN ONE ELEMENT.

71. WHAT ARE THE CHARACTERISTICS OF CERAMIC MATERIALS?

i

. THEY ARE ABRASION RESISTANT, HEAT RESISTANT (REFRACTORY) AND CAN SUSTAIN LARGE COMPRESSION LOADS EVEN AT HIGH TEMPERATURES

ii

. MANY CERAMICS ARE CHEMICALLY INERT EVEN AT HIGH TEMPERATURES AS IS SHOWN BY THEIR GOOD OXIDATION AND REDUCTION RESISTANCE AT THESE TEMPERATURES

i

ii

. CERAMICS ARE INORGANIC, NON-METALLIC MATERIALS THAT ARE PROCESSED AND/OR USED AT HIGH TEMPERATURES

AND ARE COMMONLY ELECTRICAL AND THERMAL INSULATORS,
COMPOSED OF MORE THAN ONE ELEMENT

i

v. THEY HAVE BEEN SUBJECTED HEAT TREATMENT

72.LI

STOUT THE VARIOUS TYPES OF CERAMIC MATERIALS

ACCORDING TO THEIR METHOD OF PRODUCTION?

i. WHITEWARE WHICH INCLUDES CHINA AND PORCELAIN

i

i. STRUCTURAL CLAY PRODUCTS

i

i

i. GLASS AND

i

v. REFRACTORY MATERIALS, WHICH ARE CAPABLE OF

WITHSTANDING VERY HIGH TEMPERATURES

73.LI

STHE EXAMPLES OF CERAMIC MATERIALS.

i. ALL TYPES OF GLASS PRODUCTS

i

i. ABRASIVES AND SOME TYPES OF CUTTING TOOLS

i

i

i. BRICKS, TILE, SEWER AND DRAIN PIPE

i

v. MANY TYPES OF ELECTRICAL INSULATIONS,

FERRO-ELECTRICS,

FERROMAGNETIC AND SEMI CONDUCTORS

v. REFRACTORIES FOR HIGH TEMPERATURE USE

v

i. PORCELAIN ENAMELS AND REFRACTORY COATINGS FOR

METALS

vi

i. DINNER-WARE, SANITARYWARE, ARTWARE ETC.

74.LI

STHE VARIOUS CERAMIC FORMING TECHNIQUES.

i. HYDROPLASTIC FORMING

i

i. SLIPCASTING

i

i

i. DRYI NGANDFI RING

i

v. POWDERPRESSI NG

v. CEMENTI TIOUSBOND

v

i. INJECTI ONMOLDI NG

75.WHATI SGLASS?AGLASSI SATRANSARENTSI LICAPRODUCT

WHICHMAYBEAMORPHOUSORCRYSTALLI NEDEPENDI NGUPON

HEATTREATMENT.

76.WHATARETHEMAJORSTEPSI NPRODUCI NGGLASSPRODUCTS?

i. MELTI NGANDREFI NING

i

i. FORMI NGANDSLEEPI NG

i

i

i. HEAT- TREATMENTAND

i

v. FINISHING

77.WHATARETHEFOURMETHODUSEDTOFABRI CATEGLASS?

i. PRESSI NG

i

i. BLOWI NG

i

i

i. DRAWI NG

i

v. FIBERFARMI NG

78.LI

STOUTTHTWOHEAT- TREATMENTOFGGLASSMATERI AL.

i. ANNEALI NG

i

i. TEMPERI NG

79.LI

STHTHETYPESOFGGLASS.

i. SODA-LI MEGGLASSES

i

i. LEADGLASSES

i

i

i. BOROSI LICATEGLASSES

i

v. ALUMI NOSI LICATEGLASSES

v. FUSED- SILI

CAGLASSES

80.WHATARETHEUSESOFGLASS?

i

. INDOORS, WI NDOWSANDFURNI TURES

ii

. LABORATORYEQUI PMENTSANDCHEMI CALGLASSWARE

i

ii

. DOMESTI CUTENSI LS

i

v. OPTI CALGLASSES

v. RADI OANDT. VVALVES

vi

. GLASSCURTAI NWALLONHI GHRISEBUI LDI

NG

vii

. ELECTRI CALI NSULATORS

vi

ii

. GLASSPI PE,FIBREGLASSI NSULATOR, GLASSSPRINGS

i

x. ASTHERMALI NSULATOR

x. X- RAYTUBE

81.WHATAREREFACTORI ES?

REFACTORI ESARECERAMI CMATERIALSTHATCANWI THSTAND

USUALLYHI GHHEATASWELLASABRASI ONANDTHECORROSI VE

EFFECTSOFACI DSANDALKALI S.

82.WHATARETHEUSESOFREFRACTORI ES?

i

. THEYAREUSEDI NFURNACE

ii

. FORTHECRACKI NGOFPETROLEUM, CERAMI CKILLS,BOILER,

I

NCI NERATORSETC.

i

ii

. JETENGI NESUSESREFACTORYPARTSWI THOUTIT,JET

ENGINESNOTLASTVERYLONG.

83.WHATARETHECHARACTERI STI

CSORPROPERTI ESOF

REFACTORI ES?

i

. REFACTORI ESAREHEAT, CORROSIONANDABRASI ON
RESISTANT

ii

. THEYPOSSESSHI GHFUSIONTEMPERATURE

i

ii

. THEYPOSSESSLOW THERMALCOEFFI CIENTOFEEXPANSI ON

i

v. THEYDONOTHAVECHEMI CALAFFINITYWI THTHEMOLTEN
METALTHEYHOLD

v. THEYWI THSTANDHI GHTEMPERATURESANDPRESSURE

vi

. THEYSHOULDPOSSESSHEATANDELECTRI CI

NSULATI NG

PROPERTI ES

84.THECLASSI FI

CATI ONOFREFRACTORI ESARE

i

. ACI DREFRACTORI ES

ii

. BASI CREFRCTORI ES

i

ii

. NEUTRALREFRACTORI ES

85.THEI NDUSTRI ALCONSUMERSOFREFRACTORI ESARE

i

. I RONANDSTEEL

ii

. NONFERROUSMETALS

i

ii

. STEAM ANDPOWER

i

v. OI LREFINING

v. GLASSANDPOTTERY

vi

. CEMENTLI NEANDOTHERMI NERALS

vii

. CHEMICAL INDUSTRIES

86. WHAT IS AN ABRASIVE MATERIAL? AN ABRASIVE IS A HAND, MECHANICALLY RESISTANT MATERIAL USED FOR GRINDING CUTTING.

87. LI

THE TWO TYPES OF ABRASIVE MATERIALS.

i. NATURAL (DIAMOND, GARNET, FLINT ETC.)

i

ii. SYNTHETIC (SILICON CARBIDE AND ALUMINIUM OXIDE)

88. WHAT IS CEMENT? CEMENT IS A SUBSTANCE USED AS BINDER, FOR CONSTRUCTION OF BRICKS, SETS, HARDENS AND ADHERES TO OTHER MATERIALS.

89. LI

THE TYPES OF CEMENT.

i

. NATURAL CEMENT

ii

. ARTIFICIAL CEMENT (PORTLAND CEMENT)

i

ii

. QUICK SETTING CEMENT

i

v. RAPID HARDENING CEMENT

v. SULPHATE RESISTING CEMENT

vi

. LOW HEAT CEMENT

vii

. HIGH ALUMINA CEMENT

vi

ii

. COLOURED CEMENT

i

x. BLAST FURNACE SLAG CEMENT

x. HYDROGRAPHIC CEMENT

90. LI

THE TWO TYPES OF POLYMERS.

i. NATURAL POLYMER

i

i. SYNTHETIC POLYMER

91. WHAT ARE THE CHARACTERISTICS OF POLYMERS?

i. LOW DENSITY

i

i. LOW COEFFICIENT OF FRICTIONS

i

i

i. GOOD MOULDABILITY

i

v. CAN BE PRODUCED WITH CLOSED DIMENSIONS

ON TOLERANCES

v. EXCELLENT SURFACE FINISH CAN BE OBTAINED

v

i. POOR TENSILE STRENGTH

vi

i. POOR TEMPERATURE STRENGTH

92. WHAT IS POLYMERIZATION?

i

IT IS THE PROCESS OF CONVERTING MONOMERS OR AMIXTURE OF

MONOMERS INTO A POLYMER.

93. WHAT ARE THE THREE GENERAL METHODS OR MECHANISMS OF

POLYMERIZATION?

i. ADDITION

ON POLYMERIZATION

i

i. COPOLYMERIZATION

i

i

i. CONDENSATION POLYMERIZATION

94. WHAT IS A PLASTIC MATERIAL?

PLASTICS ARE THE FINAL PRODUCT, CREATED AFTER ONE OR MORE

POLYMERS OR ADDITIVES

WES HAVE BEEN ADDED TO A RESIN DURING

PROCESSING, WHICH IT THEN SHAPES INTO A FINAL FORM.

95. LIST

THE TYPES OF PLASTICS.

i. THERMOSETTING

i

i. THERMOPLASTICS

96. WHAT ARE THE PROPERTIES OF PLASTICS?

i. THEY ARE LESS BRITTLE THAN GLASS

i

i. THEY RESIST CORROSION AND ACT AS A CHEMICAL

i

i

i. THEY ARE LIGHTWEIGHT AND POSSESS GOOD STRENGTH

i

v. THEY HAVE HIGH DIELECTRIC STRENGTH HENCE SUITABLE FOR ELECTRIC INSULATION

97. THE USES OF PLASTICS ARE.

i. IT IS USED FOR MAKING DISPOSABLE COMMODITY PLASTICS

i

i. IT CAN BE USED TO REPLACE METALS IN SOME PARTS OF MACHINES

i

i

i. IT CAN BE USED TO MAKE INSULATOR

98. WHAT IS THE DIFFERENCE BETWEEN THERMOSETTING AND THERMOPLASTIC

THERMOSETTING RESINS THERMOPLASTIC

i

. ONCE HARDENED AND THEY CAN BE REPEATEDLY SET, THEY DO NOT SOFTEN BY HEAT AND SOFTEN WITH THE HARDENED BY COOLING

APPLICATIONS OF HEAT.

i

i

. THEY ARE STRONGER THEY ARE COMPARATIVELY AND HARDER THAN SOFTER AND LESS STRONG THERMOPLASTIC

RESINS

OBJECTS MADE BY THERMOPLASTIC RESINS

i

i

i. OBJECTS MADE BY CANNOT BE USED AT

THERMOSETTI
 NG
 COMPARATIVELY HIGHER
 RESINS CAN BE USED AT TEMPERATURES AS THEY WILL
 AT COMPARATIVELY TEND TO SOFTEN UNDER HEAT
 HIGHER
 TEMPERATURE
 WITHOUT DAMAGE.

i
 v. THEY ARE USUALLY SUPPLIED AS
 EDAS
 SUPPLIED IN GRANULAR MATERIAL.
 MONOMER OR
 PARTIALLY
 POLYMERISED FORM
 I
 N WHICH THEY ARE
 EITHER
 THERMALLY CURED OR
 PARTIALLY
 THERMOPLASTIC
 SOLIDS

99. THERMOSETTING PLASTICS CAN BE FORMED BY.

i. COMPRESSION OR TRANSFER MOLDING

i

i. CASTING

100. THERMOPLASTICS CAN BE FORMED BY.

i. INJECTION MOLDING

i

i. BLOW MOLDING

i

i

i. EXTRUSION

i

v. CALENDERING

v. THERMOFORMING

v

i. CASTING

101. WHAT ARE ELASTOMERS (RUBBERS)

AN ELASTOMER IS A POLYMERIC MATERIAL THAT MAY EXPERIENCE LARGE AND REVERSIBLE ELASTIC DEFORMATIONS.

102. WHAT ARE THE CHARACTERISTICS OF RUBBERS?

i. THEY ARE ESSENTIALLY NON-CRYSTALLINE IN STRUCTURE

i

i. THEY ARE NON-CONDUCTORS OF ELECTRICITY AND ARE RELATIVELY LOW HEAT CONDUCTORS.

i

i

i. THEY HAVE RELATIVELY LOW SOFTENING TEMPERATURES

i

v. THEY ARE HIGHLY RESISTANT TO CHEMICALS AND RESISTANT TO CHEMICAL CORROSION IN ENVIRONMENT

v. THEY EXHIBIT

SCHEMATIC BEHAVIOUR GENERALLY TO A

GREATER EXTENT THAN PLASTICS.

103. WHAT ARE THE PROPERTIES OF ELASTOMERS?

i

i. RESILIENCE AND HYSTERESIS

i

i

i. HARDNESS

ii

i

i. TENSILE STRENGTH

iv

i. COMPRESSION SET OR CREEP

v

i. TEAR RESISTANCE

vi

i. ABRASION RESISTANCE

vii

i

i. DETERRIORATION RESISTANCE

viii

i

i. OIL RESISTANCE

104. LIST THE TYPES OF ELASTOMERS OR RUBBERS

i

i. NATURAL RUBBER

i

i

. STYRENEBUTADIENE RUBBER(SBR)

ii

i

. CHLOROPHENE(NEOPRENE)(CR)

iv

. ISOBUTYLENE- ISOPRENE(BUTYL)(UR)

v

. ISOPRENE

vi

. POLYACRYLATE(ACM)

v i

i

. NITRILE BUTADIENE(NBR)

vii

i

. POLYBUTADIENE(BR)

i

x. POLYSULFIDE(THIOLCOL)

x. ETHYLENEPROPYLENE(EPDM AND EPDM)

xi

. URETHANE(AUANDEU)

xii

. CHLOROSULFONOLYMPOLYETHYLENE(HYPALON)(
CSM)

FLUOROCARBON(FKM)

xii

i

. SILICONE

105. WHAT IS VULCANIZATION OF RUBBER?

VULCANIZATION IS THE REVERSIBLE CHEMICAL REACTION
ON

I

INVOLVING SULPHUR OR OTHERS IN THE PRESENCE OF
CROSSLINKS ARE FORMED BETWEEN MOLECULAR CHAINS IN RUBBER
MATERIALS.106. LIST THE FORMING AND FABRICATING TECHNIQUES FOR
RUBBERS

i

. COMPRESSION MOLDING

i

i

. CALENDERING

ii

i

. EXTRUSION

iv

. INJECTION AND TRANSFER MOULDING

v

. VACUUM FORMING OF SHEET RUBBER

vi

. CASTING OF LIQUID (URETHANE AND POLYSULPHIDE) RUBBERS

107. THE FINEST METHODS FOR RUBBERS ARE

i. TUMBLING, EITHER DRY OR IN LIQUID CARBON-DIOXIDE

TO

REMOVE FLASH

i

i. DIE TRIMMING

ii

i. TEAR TRIMMING

iv. BUFFING

108. WHAT ARE THE USES OF RUBBER?

i

. VEHICLE TYRES

i

i

. ERASERS

ii

i

. GASKETS

iv

. GLOVES AND APRONS

v

. BELT CONVEYORS

vi

. SEALS

vi

i

. O-RINGS

vii

i

. WATERSTOPS

i

x. HOSES

x. FLOORTI LES

xi

. ADHESI VESANDTAPES

xii

. EXPANSI ONJOI NTS

xii

i

. CORROSI ONRESI STANTCOATI NGS,LINNI NGS

xiv

. PI PESANDCONTAI NERS

xv. WI REANDCABLECOVERI NGS

xvi

. VI BRATIONMOUNTSI NAUTOMOBILE

xv i

i

. AUTOMOBILEUPHOLSTERY

xvii

i

. ENGINEMOUNTI NGS

109.WHATARECOMPOSITEMATERIALS?

AREMADEOFDIFFERENTMATERIALSTHATAREFORCED

TOGETHER(EXAMPLE: FIBERGLASS, CONCRETE, ANDWOOD)TO

ACHIEVESPECIFICPROPERTIES.

110.LISTTHETYPESOFFIBER-REINFORCEDCOMPOSITES.

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i

. SYNTHETICAND

i

i

. NATURAL

111.WHATARETHEUSESOFFIBER-REINFORCEDCOMPOSITE?

111.WHATARETHEUSESOFFIBER-REINFORCEDCOMPOSITE?

i

. THEYAREUSEDTOREPLACEALUMINIUMALLOY,WHICHPROVIDE

HIGHSTRENGTHANDFAIRLYHIGHSTIFFNESSATLOWWEIGHT.

i

i

. ARE USE FOR STRUCTURAL APPLICATIONS

121. WHAT ARE THE PROPERTIES OF FIBER-REINFORCED COMPOSITE

i

. THEY HAVE HIGH TENSILE STIFFNESS, AND STRENGTH

i

i

. HAVE VERY SMALL BENDING STIFFNESS

ii

i

. THEY ARE HARD BUT BRITTLE

122. WHAT PROPERTIES OF COMPOSITES DEPEND ON?

i. PROPERTIES OF PHASES

i

i. GEOMETRY OF DISPERSED PHASE (

PARTI

CLES

ZE,

DI

STRIBUTION,

AND ORIENTATION

ON)

i

i

i. AMOUNT OF PHASE.